Revenue Architecture: Unit Economics, Financial Model, and Pricing Test

Course	BUS 395: Venture Creation with AI — Rollins College — Summer 2026
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Date	June 2026 (Week 5)
Venture	Pre-Diabetic Family Weekly Food System
Revenue Model	Grocery chain partnership (chain pays per active user; end user pays $0)

1. Revenue Model Selection

Models Eliminated

Direct subscription (<$5/month). The customer's discretionary spend on food-planning tools is functionally zero. Grace has a hard $60 weekly grocery ceiling; Nelsy canceled a meal kit on price; Veronica abandoned a *free* app over friction. Competitive research shows $5–15/month is already a structural barrier before groceries. Pricing below that barrier with no willingness-to-pay data is a guess, not a model.

One-time purchase. The product's value is weekly regeneration — new budget, new sales circular, new plan every week. A one-time purchase does not match a product whose core value is ongoing. No interviewee showed any signal of paying once for a recurring problem.

Freemium. Freemium only generates revenue if free users convert to paid. This segment converts at near-zero rates because of income constraints. Veronica proves a free tier still fails on friction, so the free tier would have to be frictionless — which makes the paid upgrade even harder to justify. You would be building a free product for an audience that cannot upgrade.

Transaction fee (per plan). Pay-per-use aligns with inconsistent engagement, but inconsistent use means inconsistent revenue, and it trains the customer to skip weeks — the opposite of the habit the product needs to build.

B2B clinic / employer ($30–50/member/month). The strongest financial logic of any option: one patient progressing to Type 2 diabetes costs a payer $9,600–16,000/year, so a prevention tool has asymmetric ROI. But there is zero interview validation — not one interviewee named a doctor, clinic, or employer as a food resource. PlateJoy validated this institutional concept and still failed: routing through insurance added HIPAA friction and tied revenue to health outcomes it could not control. Sales cycles run 6–12 months with no existing relationships. Kept as a future layer, not the entry model.

Model Selected: Grocery Chain Partnership

A regional grocery chain pays a per-user licensing fee for access to a planning tool that builds each pre-diabetic shopper's weekly meals around that chain's sales circular and the shopper's budget. The end user pays nothing.

How money moves: The chain funds the tool as a shopper-engagement / loyalty benefit (marketing budget, not IT procurement). The shopper gets a free weekly plan and grocery list built around the current circular. The chain pays a monthly per-active-user fee. Working estimate: **$5/active user/month** — below PlateJoy's $30–50 healthcare rate (no HIPAA burden), above basic $1–2 loyalty punch-card tech.

Evidence for:

- Every interviewee went to the store without a plan and *put back a healthier item because they did not know what meal to make with it.* Tere walked away from healthier food that "felt expensive and I didn't know what meals to make with it." That exact failure is the chain's revenue problem, not just the customer's health problem.
- A shopper who arrives with a plan buys intentionally, wastes less, and returns on a reliable weekly cycle — basket-size lift plus loyalty plus retention, the three things a chain's marketing team is already measured on.
- Grace's $60 ceiling is evidence *for* this model: a chain captures a budget-constrained shopper who was already

going to spend $60 and turns her into a loyalty customer. The user gets a free system; the chain captures basket quality.

- It sidesteps the price-sensitivity problem entirely. The user pays nothing, so the documented inability to pay $5–15/month stops being a barrier.

Evidence against:

- No grocery chain partner has been identified. There is zero interview data from the chain's side.
- Grocery retail moves slowly; a pilot deal likely exceeds the 12-week class timeline. This model requires a B2B sale before the product exists at scale.
- The friction problem (Veronica's free-app abandonment) still applies on the user side — that is a product-design risk the partnership does not solve by itself.

> **Why this beats the alternatives:** It is the only model where the payer's financial incentive directly matches the customer's documented behavioral failure. The customer cannot pay; the chain can, and benefits when she shops with a plan.

2. Unit Economics

Full unit economics with every assumption labeled are in the attached Excel (Unit Economics sheet). Below is the analysis. Every number is tagged **[DATA]** (grounded in API pricing, published competitive research, or interview evidence) or **[ASSUMPTION]** (a reasoned estimate that must be tested).

Revenue

The unit of revenue is one active user per month. Working estimate: **$5.00/user/month [ASSUMPTION]**. No chain has been approached, so this has no real-world validation. It is bracketed by two anchors: PlateJoy's $30–50/member/month healthcare rate [DATA] above, and $1–2 basic loyalty tech below.

Costs (the audit that mattered)

The first cost pass produced a 95% gross margin. That was a red flag, not a result — the lab warns that any margin above 80% means costs are missing. After pushback, three costs were corrected and five were added:

- **AI inference** was 3× too low. A real plan needs the system prompt with blood-sugar rules, family profile, pantry, the grocery circular (~3,000 tokens alone), and last week's plan to prevent repetition — ~6,900 input tokens, not 3,000 — plus mid-week interactions. Revised to **$0.35/user/month [DATA, from published Claude API pricing]**.
- **Clinical advisor** was scoped for launch, not operation. 8 hrs/month reviews under 1.5% of plans at 500 users. Revised to 20 hrs/month at $150/hr = **$3,000/month [ASSUMPTION]**.
- **Insurance** was budgeted as a general business. A product making dietary recommendations for a diagnosed condition needs E&O + product liability + cyber/data-breach = **$600/month [ASSUMPTION]**.
- **Added entirely:** data-integration maintenance ($400/mo), compliance tooling ($200/mo), customer support ($0.15/user), QA automation ($100/mo), and a security-grade hosting tier ($200/mo base).

Margins

After the audit, variable cost is **$0.60/user/month** and gross margin is **$3.20/user (64%)** at the $5 rate — squarely in the realistic 60–70% range for AI-native, health-adjacent SaaS. At the $2 low rate, margin collapses to 10% and the product loses money on every user before fixed costs apply at all — clear evidence $2/user is not a viable contract price.

Metric	First pass	After audit	What changed
AI inference / user / mo	$0.17	$0.35	Token count was 3× too low
Variable cost / user / mo	$0.27	$0.60	Added customer support; revised inference
Fixed costs / month	$1,250	$4,500	Clinical advisor, insurance, integration, compliance
Gross margin / user	$4.73 (95%)	$3.20 (64%)	Hosting + integration moved into COGS
Break-even active users	264	~1,200	4.5× higher

Break-even

At $5/user and $3.20 gross margin, break-even is **~1,200 active users** from a single chain deal — roughly 2.4% activation of a 50,000-member loyalty base. Achievable, but only if the chain actively promotes the tool. That promotion commitment is a contract term, not an afterthought.

3. Financial Model: First 6 Months

The full model with three scenarios and the assumptions column is in the attached Excel (Financial Model sheet). The three scenarios differ by *assumptions* — when the deal closes, whether the chain promotes the tool, and the churn rate — not just by volume. Summary:

Scenario	Key assumptions	Deal closes	Churn	6-mo net
Pessimistic	Cold outreach, no champion; chain treats tool as background feature	Month 6	40%/mo	(−$25,900)
Base	Deal closes mid-window; modest promotion; 0.5% activation	Month 3	20%/mo	(−$16,572)
Optimistic	Warm intro; chain actively promotes; 2% activation	Month 2	10%/mo	approaching break-even

What This Tells Me

- **No scenario breaks even in 6 months.** The base case ends at −$16,572, still heading toward break-even. That is the honest output of a real stress test, not a failure.
- **The single most important short-term variable is when the deal closes.** Each pre-revenue month costs $4,500 with zero income. Moving a close from Month 4 to Month 2 saves $9,000 without changing anything else.
- **The most important long-term variable is the chain rate.** $2 vs. $5 is a binary viability gate: at $2/user, break-even needs ~3,214 users from one regional pilot — not achievable. At $5, it is ~1,023 — hard but real. You cannot negotiate into viability from $2; you have to decline that structure.
- **What would have to change to break even:** defer the $4,500/month operating cost stack until a deal is near (pre-launch real cost is ~$500–800/month), secure active chain promotion to push activation toward 2%, and — most important — solve churn.

4. Pricing Test Plan

Important framing

The payer (the grocery chain) is not the interviewee. So this week's test cannot ask the chain anything — it tests the *user* side: will a complete, store-specific weekly plan actually change what these shoppers buy? That behavioral evidence is the proof I would later bring to a chain. It also produces a direct-to-consumer fallback price signal.

Hypothesis

"Tere, Grace, and Mary will **use** a free, complete, store-specific weekly plan built around their budget and this week's grocery sales — changing what they actually buy — because all three already shop without a plan and put healthy items back for lack of a meal idea."

Test Method: Wizard of Oz

Manually build (behind the scenes) a real weekly plan for each person around a real store circular and their stated budget. Deliver it the way they already communicate (text / WhatsApp). Watch whether they follow it — behavioral evidence, not opinion. This is the right method because the entire venture hinges on whether a complete output bridges to action, which Nelsy's mother's-recipes pattern says is the real question.

Step-by-Step Plan

- **Sat–Sun:** Text Tere, Grace, and Mary. Confirm their weekly budget, household size, and which store they shop. Pull that store's current circular.
- **Sun–Mon:** Hand-build each person a 5–7 day blood-sugar-aware plan + grocery list priced to their budget, built around this week's sales.
- **Mon:** Send each plan before their next shopping trip.
- **After their trip:** Ask what they actually bought and cooked, and whether they want next week's plan. As a DTC fallback price signal: "If this came every week, what would you pay for it?"
- **Fri:** Write up per-person results — what they bought, what they cooked, whether they asked for week 2.

Success vs. Failure Signals

Signal	What it looks like
Yes	At least 2 of 3 change what they buy based on the plan, and at least 1 asks for next week's plan unprompted. The 20% churn assumption gains behavioral support; the venture has a viable path.
No	They read it, call it helpful, and buy the same cart anyway — the Nelsy pattern. Complete, ready-to-use info still does not bridge to action.

If the Test Fails (Plan B)

- The churn problem is not solvable by better plan delivery alone. Shift the intervention from "deliver a plan" to "reduce execution effort" — e.g., pre-built cart the store can fulfill, or a single-tap reorder, testing whether removing the last execution step changes behavior.
- Re-test with a smaller, sharper output (one or two anchor meals around the week's best sale item) rather than a full plan, to see if a lower-effort ask converts.
- If neither moves behavior, the finding itself is the deliverable: this customer needs a fundamentally different product intervention before any revenue model is worth building.

5. Updated VENTURE.md: Revenue Model Section

Chosen model: Grocery chain partnership. A regional grocery chain pays a per-user licensing fee for access to pre-diabetic shoppers who plan their weekly meals around the chain's sales circular. The end user pays nothing. Working price estimate: $5/active user/month, positioned below PlateJoy's $30–50 healthcare rate. Unvalidated — no chain has been approached.

Unit economics: At $5/user/month, variable costs run $0.60/user (AI inference, hosting, support) for a 64% gross margin once hosting and data integration are in COGS. Fixed operating costs (clinical advisor 20 hrs/mo, insurance, compliance, data-integration maintenance) total $4,500/month. Break-even is ~1,200 active users, roughly 2.4% activation of a 50,000-member loyalty base. The swing variable is whether the chain actively promotes the tool, not the per-user rate.

Biggest financial risk: Churn. All five interviewees abandoned every tool they tried — a free app, a meal kit, a cookbook, a diabetes app, TikTok, and a trusted family member's personalized recipes. The base scenario's 20% monthly churn is a hope; the interview data supports 40%. At 40%, a 250-user base decays to 11 within six months and the model never recovers. Viability depends on solving the abandonment problem every prior tool failed to solve.

What the pricing test must resolve: Whether a complete, store-specific weekly plan actually changes what Tere, Grace, and Mary buy. If 2 of 3 change their cart and 1 asks for another, the retention assumption has support. If none follow it, the product replicates the Nelsy pattern and the revenue model needs a different product intervention before any projection is worth building.